

Project Proposal: Recovering Cyber-Security Domain Knowledge via LoRA Fine-Tuning

Team Members: Farit Sharafutdinov, Grigorii Belayev

Course: Generative AI (Spring 2026)

Instructor: Bader Rasheed

Problem Statement & Motivation

Modern Large Language Models (LLMs) undergo extensive safety alignment to prevent the generation of harmful content. However, this often leads to "over-refusal," where models reject legitimate cyber-security tasks such as vulnerability research, code auditing, or defensive strategy explanation. This project aims to investigate whether LoRA (Low-Rank Adaptation) can effectively recover this suppressed domain knowledge and reduce false refusal rates without compromising the model's fundamental safety guardrails.

Data

- **Source:** A curated dataset combining the **Cyber-Security-Instruct** set (based on CTF challenges and CVE reports) and filtered technical documentation.
- **License:** Open-source data under Apache 2.0 or MIT licenses.
- **Preprocessing:** Data will be converted into Instruction-Response pairs, deduplicated, and cleaned for technical accuracy.

Baseline Plan

The baseline will be the **Llama-3-8B-Instruct** model. We will evaluate its out-of-the-box performance on 50 complex cyber-security queries to establish a "Refusal Rate" and "Helpfulness Score" using standard system prompts.

Evaluation Plan

- **Quantitative Metrics:** Refusal Rate (percentage of "I cannot help" responses) and Cyber-Accuracy (correctness of technical explanations).
- **Qualitative Analysis:** Comparison of "Before vs. After" response samples using consistent random seeds.
- **Protocol:** A controlled comparison between the baseline and the LoRA-tuned model across varying difficulty tiers.

Risks

- **Compute:** Fine-tuning 8B models may exceed standard 16GB VRAM. **Mitigation:** Utilizing 4-bit quantization (QLoRA) on Kaggle/Colab environments.
- **Catastrophic Forgetting:** The model might lose general reasoning capabilities. **Mitigation:** Monitoring performance on general benchmarks during training iterations.

Ethics & Safety Considerations

This project focuses on "White-Hat" defense restoration. To mitigate misuse risks, the model will be tested to ensure it still refuses requests for creating malicious software (e.g., ransomware) or engaging in illegal activities.

What is happening inside?

As we are adapting a large pretrained model, we provide the technical mechanism of the adaptation:

- **Model Architecture:** We apply Low-Rank Adaptation (LoRA) to the Query (W_q) and Value (W_v) weight matrices within the Self-Attention layers.
- **Training Objective:** Standard Causal Language Modeling (CLM) loss: $\mathcal{L} = -\sum_{t=1}^T \log P(x_t|x_{<t}, \Delta W)$.
- **Key Training Details:** We train low-rank matrices A and B (rank $r = 16$) while freezing the original model weights. This allows for parameter-efficient updates without modifying the backbone.

Timeline

- **W2-W3:** Dataset acquisition and final task framing.
- **W3:** Stage 1: Proposal Submission.
- **W5:** Stage 2: Baseline implementation and initial Refusal Rate testing.
- **W7-W9:** Improvement sprint: LoRA fine-tuning and rank r ablations.
- **W10-W12:** Final evaluation, code reproducibility checks, and demo preparation.

References

- [1] Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models.
- [2] Dettmers, T., et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs.
- [3] Touvron, H., et al. (2023). Llama 2/3: Open Foundation and Fine-Tuned Chat Models.